

Build A PC

The full written lesson — compatibility, assembly, BIOS/UEFI, OS install/thermals/troubleshooting.

1. Compatibility: the part that actually matters most

Nearly every build problem traces back to a compatibility decision made — or skipped — before any part was purchased.

- The **CPU socket** must match the motherboard exactly — socket names are sometimes reused across CPU generations, which creates its own trap.
- **RAM generation** (DDR4 vs DDR5) is dictated by the motherboard, not the CPU alone.
- The **QVL** is the motherboard manufacturer's tested-compatible RAM list.
- **PSU wattage** should be sized with headroom — a GPU can briefly draw well above its rated TDP under load.

The trap: assuming "same socket name" always means compatible. An old motherboard's firmware may not recognize a newer CPU until flashed — sometimes needing an older compatible CPU or a BIOS-flashback feature just to update first.

Worked: search the motherboard's official QVL for your exact RAM kit before buying — real evidence, not marketing compatibility claims.

2. Assembly: static safety and the order that saves you time

Assembly mistakes are almost all preventable with two habits: grounding yourself, and testing before you commit to the case.

- **ESD** damage is often not immediately obvious — an anti-static strap removes the risk for the cost of \$7.
- CPU sockets are **zero-force-insertion** — resistance means stop and recheck alignment.
- **Test-boot outside the case first** to isolate component problems from assembly problems.

The trap — the single most common first-build mistake: forgetting the separate CPU power connector (EPS). The system can power on — fans spin, lights on — without it, but fails to POST, looking exactly like a mysterious dead board.

Worked: test-booting outside the case catches the EPS-cable mistake in under a minute, versus discovering it after the case is fully closed up.

3. BIOS/UEFI: settings that matter before an OS exists

UEFI configures hardware before any OS is involved — and hides one setting that quietly undersells your own RAM if you skip it.

- **Boot order** must point at installation media first when installing an OS.
- **XMP** runs RAM at its actual rated speed — without it, RAM defaults to a conservative speed well below what's printed on the box.

The trap: over-tightening the cooler mount, or applying far too much thermal paste. A pea-sized amount is typically sufficient — excess doesn't help and can cause real problems.

Worked: RAM rated 6000MT/s showing as 2133MT/s isn't a hardware fault — it's XMP never having been enabled.

4. OS install, thermals, and diagnosing a dead build

The last mile — getting an OS on the machine — and the systematic process for when something earlier doesn't work.

- **Bootable media** (Rufus/Ventoy) writes an OS installer to USB.
- **Airflow** needs a coherent path — insufficient airflow causes **thermal throttling**.
- **POST troubleshooting order:** power connections → RAM reseal → CPU power cable → standoffs → debug LEDs/beep codes.

The trap: buying a PSU on price alone. Protection circuitry matters more than the wattage number, and it's the one component whose failure can take others down with it.

Worked — the systematic order in action: no display, no beep. Power confirmed present. RAM reseated — no change. CPU power cable rechecked — already seated. Standoffs checked — one found shorting. Removed, system POSTs. Five ordered checks, no guessing.